

SAHIL RAI

Delta, BC | raisahil580@gmail.com | (236) 858-9811

WORK EXPERIENCE

Backend Engineer (Full-time) Dec 2023 - Present

FusionWare Integration Corp.

- Developed and continues to enhance 20+ microservices, CDN services and API Gateways using Go or Node.js (TypeScript).
- Reduced API response time by 50% through Redis caching, concurrent processing and Kafka-based event-driven architecture.
- Utilized Docker and Kubernetes for container orchestrations and Jenkins for CI/CD automation in development workflow.

Full Stack Developer (Part-time, Contract) Mar 2024 - Jul 2024

Risings Education Inc.

- Developed a full-stack project management app using Mongo, TypeScript, Node.js, Redis, and OAuth2 in the backend.
- Used React (TypeScript) and Tailwind CSS for frontend, deploying the frontend, backend, and Redis on AWS EC2.
- Managed and optimized AWS infrastructure, including Lightsail, EC2, Lambda, and CloudFront.

Software Engineer (Full-time) Jan 2023 - Dec 2023

Buble

- Worked in a team of 5 to develop a web-based audio chat app using React and Node.js (TypeScript) for 10,000+ users.
- Built a scalable backend to handle 1,000+ concurrent requests, utilizing PostgreSQL and AWS S3 for data storage.
- Created detailed API documentation, collaborated in an Agile development environment with Scrum and CI/CD pipelines.

Web Developer Intern (Full-time) Jan 2022 - Aug 2022

Accenture

- Collaborated on MERN projects, enhanced UI and worked on Node.js for backend management and routing.
- Reduced page load time by 40% through lazy loading and code splitting, improving user retention.

PUBLICATION

Shvetha S Kumar, Reshma R Nayak, Jismi S Kannampuzha, **Sahil Rai**, Jeeho Ryoo, and Lizy K John, "[Evaluation of Pruning Techniques](#)", Proceedings of the *IEEE International Performance, Computing and Communication Performance (IPCCC)*, Anaheim, CA, USA, November 2023

RESEARCH

Lead Researcher - Backend Security (Company-Sponsored) Jan 2025 - Present

Internal R&D, FusionWare Integration Corp.

- Developing a novel API security framework using fully homomorphic encryption for computations on encrypted data.
- Employed Zero-Knowledge Proofs for authentication, eliminating passwords and preventing server data exposure.
- Implementing Node.js backend to demonstrate practical deployment in high-security, covert communication systems.

Research Assistant - Machine Learning (Academic) Jan 2023 - Mar 2023

Centre of Applied Research, BCIT

- Investigated Pruning Techniques in Neural Networks.
- Analyzed L1-Norm, network slimming, and weight-level pruning with fine-tuned models and diverse datasets.
- Explored CNN models like ResNet(s) and VGG-16 using TensorFlow with optimized hyperparameters.

OPEN SOURCE CONTRIBUTIONS

- Detekt Kotlin Code Reviewer (JUnit, coroutines, REST API)** Contributed to Kotlin repository by ensuring the enforcement of runTest in JUnit when using coroutines. Enhanced reliability in asynchronous testing for more robust scenarios. Tech: Kotlin, JavaScript, Kafka
- SAR Image Recognition Site (Concurrent and Batch Processing, API Optimization)** Enhanced search and rescue web app, cutting image processing time by 65% (35 mins to 12 mins from 5K images) via parallel processing. Implemented personalized search and concurrent API result display. Tech: Node.js, React, WebSocket, TypeScript, Mongo

LINKS

[LinkedIn.com/in/Rai-Sahil](https://www.linkedin.com/in/Rai-Sahil)

[GitHub.com/Rai-Sahil](https://github.com/Rai-Sahil)

[Twitter.com/imsahil931](https://twitter.com/imsahil931)

EDUCATION

British Columbia Institute of Technology

BASc in Computer Systems

Dec 2021 | Burnaby, BC

Major: Software Engineering

COURSEWORK

Machine Learning
Software Engineering
Web Development
Backend Technologies
System Design
Operating System
Cloud Computing
Data Communication
Discrete Maths
Data Structures and Algorithms

SKILLS

Languages:

JavaScript, TypeScript, C#, Golang, Python

Framework:

Node.js, React, Next.js, .NET, Tailwind CSS, Gin, gRPC
GraphQL, Django, ASP.NET, REST

Database:

PostgreSQL, MySQL, MongoDB, Redis, SQLite
ElasticSearch, MS SQL Server

DevOps/Infrastructure:

Kubernetes, Docker, Jenkins, Azure DevOps, GORM
Apache Kafka, S3, Lambda, CloudFront, EKS

CERTIFICATES

Connect & Protect: Networks (Google, 2023)

Algorithms, Part II (Coursera, 2023)

React Essential Training (LinkedIn, 2021)